Sundram Kumar Rai

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EDUCATION

National Institute of Technology Raipur — **Aug 2023 - May 2027**
Bachelor of Technology, Electrical Engineering | ***CGPA: 8.86 / 10.0*** — *Raipur, Chhattisgarh*

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript
CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, Concurrency
Development: React, Node.js, Express.js, FastAPI, REST APIs, WebSockets, MongoDB, PostgreSQL, Redis
AI / GenAI: RAG, LLMs, Generative AI, AI Agents, pgvector, Structured Outputs
Testing & DevOps: Azure App Service, Docker, GitHub Actions, CI/CD, Git, Linux, Postman, Jest

WORK EXPERIENCE

Project Lead — PeopleOps Copilot — **May 2026 – Aug 2026**
Technologies: Python, FastAPI, LangGraph, MCP, PostgreSQL, RAG, Docker — *Innovation Cell, NIT Raipur*

- Built an **agentic employee assistant** with **LangGraph, FastAPI, MCP** for HR policy and workforce workflows.
- Implemented **NL-to-SQL** analytics with **RBAC, query validation, and HITL approval** for sensitive actions.
- Added **audit logging, structured outputs, and PostgreSQL** for traceable, policy-grounded agent execution.

PROJECTS

ForkSpace | GitHub | React, Socket.IO, Node.js, Express, MongoDB, Redis — **Mar 2026**

- Architected **ForkSpace**, a real-time collaborative code editor with **500+ visits**, enabling multi-user editing and code execution via **CodeMirror, Socket.IO**, and **Judge0 API**.
- Implemented **RBAC**, **automated hidden-test validation**, deterministic stdin generation, and **Piston** for multi-language execution; deployed on **Azure App Service** with **Docker**.
- Engineered a **Redis token-bucket** rate limiter with atomic **Lua** scripts to prevent concurrent race conditions, achieving **17.3 ms** edit latency.

ReXcan | GitHub | FastAPI, OCR, LLMs, Node.js, React, Python — **Nov 2025**

- Built **ReXcan** to automate invoice extraction and validation using **FastAPI**, **pdfplumber, EasyOCR, Tesseract**, and **LLM** fallback, achieving **86% accuracy** and **2.56 invoices/sec** throughput.
- Implemented **duplicate detection, arithmetic validation**, and audit trails using a **BullMQ + Redis** asynchronous workflow to ensure reliable, repeatable processing.
- Created a human-in-the-loop React.js review interface with **Jest unit-tested** correction workflows for manual verification.

RecallLoop | GitHub | React, TypeScript, Node.js, PostgreSQL, RAG — **June 2026**

- Developed **RecallLoop**, an adaptive learning platform using **React, Node.js**, and **PostgreSQL**, modeling learner mastery and goals to generate personalized recall, planning, and review schedules with **FSRS**.
- Engineered an asynchronous **Generative AI pipeline** using **Redis-backed queues, RAG, and pgvector** for concept extraction and rubric-based evaluation, with **retries, idempotency, structured outputs**, and agentic tools.

ACHIEVEMENTS

- Selected for the **Amazon ML Summer School 2026** from over **1.25 lakh** applicants.
- **Codeforces**: Specialist (1406) — **CodeChef**: 3-Star (1731)
- **LeetCode**: **Knight** (1904), Global Rank **869** among **30,390 participants** in Weekly Contest 469
- Solved **1,500+ DSA problems** across multiple competitive programming platforms.
- Awarded **INR 25,000** seed grant for an ML-based pet health startup (smart collar with anomaly detection).
- **Open-Source Contributor** — Merged **10+ PRs** in open-source repositories, contributing to **CI/CD, testing, accessibility, and frontend features** through **GSSoC '25** and **Hacktoberfest '25**.

LEADERSHIP EXPERIENCE

- Technical Lead, Robotix Club, NIT Raipur (**Sep 2023 – Dec 2025**): Led software development for autonomous robotics projects, implementing **A* pathfinding in Python** and mentoring **50+ students** in algorithm design and debugging.